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EX.NO:01

DATE:

**IDENTIFY A SOFTWARE SYSTEM THAT NEEDS TO BE
DEVELOPED**

AIM:

To Identify a software system that needs to be developed.

ALGORITHM:

Step 1: Identify the problem or need that the software system aims to address. Clearly define the objectives and scope of the system, considering the stakeholders' requirements.

Step 2: Identify and involve key stakeholders who will be impacted by the software system. Gather input from end-users, managers, and any other relevant parties to understand their needs and expectations.

Step 3: Evaluate the technical, economic, operational, and schedule feasibility of developing the software system. Assess whether the proposed system is viable and justifiable given the available resources.

Step 4: Document functional and non-functional requirements based on stakeholder input. Specify features, constraints, and any other relevant information that will guide the development process.

Step 5: Prioritize requirements based on their importance and impact on achieving the system's objectives. Identify essential features that must be included in the initial development phase.

Step 6: Research existing software solutions to identify if there are pre-existing systems that can fulfill the requirements. Analyze the strengths and weaknesses of existing solutions to determine if customization or a new system is necessary.

Step 7: Create a high-level design of the software system, outlining the major components and their interactions. Consider scalability, maintainability, and other architectural considerations.

Step 8: Develop a prototype or a minimal viable product (MVP) to validate the design and gather feedback from stakeholders. Iterate on the prototype based on feedback to ensure alignment with user expectations.

Step 9: Based on feedback from the prototype, refine and finalize the system requirements. Ensure that the requirements are clear, complete, and feasible for implementation.

Step 10: Create a development plan outlining tasks, timelines, and resource requirements. Plan for testing, quality assurance, and deployment phases.

PROCEDURE:

1. Identify the Problem or Need:

Clearly articulate the problem or need that the software system is intended to solve.

2. Gather Stakeholder Input:

Engage with stakeholders to collect requirements and understand their expectations.

3. Evaluate Feasibility:

Assess the feasibility of developing the software system in terms of technical, economic, operational, and schedule aspects.

4. Create Requirements Document:

Document the requirements, specifying features, constraints, and any other relevant information.

5. Prioritize and Plan:

Prioritize requirements and create a development plan outlining the order of implementation.

6. Explore Existing Solutions:

Research existing software solutions to determine if customization or a new system is required.

7. Design System Architecture:

Develop a high-level design of the software system, considering scalability and maintainability.

8. Prototype and Validate:

Build a prototype or MVP to validate the design and gather stakeholder feedback.

9. Refine Requirements:

Refine and finalize the requirements based on feedback from the prototype.

10. Document Development Plan:

Create a detailed development plan, outlining tasks, timelines, and resource requirements.

11. Iterate and Develop:

Iterate on the development process, building and refining the software system according to the finalized requirements.

12. Test and Quality Assurance:

Conduct thorough testing to ensure the software meets the specified requirements and quality standards.

13. Deploy and Maintain:

Deploy the software system and establish maintenance procedures for ongoing support and updates.

14. Collect User Feedback:

Gather feedback from end-users after deployment to identify any potential improvements or additional features.

15. Iterate for Continuous Improvement:

Iterate on the software based on user feedback and evolving requirements to ensure continuous improvement.

RESULT:

Thus a software system was developed Successfully.